# Developmental delay & intellectual disability What Indian data shows, and the testing order doctors follow

For sales and healthcare teams. Facts summarised in our own words from published sources (links below). Treatment decisions are always the doctor's.

**Indian study:** Sharma AR, Siddiqui MS, Magar S, Kale A, Nelanuthala M, Singh SP. *The Etiological Profile of Global Developmental Delay at a Tertiary Care Hospital in India: An Observational Study.* Cureus 2023;15(6):e41066. doi:10.7759/cureus.41066. MGM Medical College and Hospital, Aurangabad. Open access (CC BY 4.0).

## Who was studied

- 52 children aged 3 months to 5 years with global developmental delay (developmental quotient below 70%)
- Cause found in about 88% (the paper reports 88.4% and 89.4% in different places)
- Tests used by clinical need: MRI, EEG, metabolic tests, karyotype, microarray, exome or genome

## What caused the delay

- **Prenatal 50%:** genetic 32.7%, chromosomal 7.7%, brain malformation 7.7%, congenital infection 1.9%
- **Perinatal 34.6%:** birth asphyxia (HIE) 26.9%, intracranial haemorrhage 7.7%
- **Postnatal 3.8%**; **unknown 11.6%**
- Exome/genome found a genetic cause in 17 children, including Tay-Sachs, Niemann-Pick A, MPS II, Gaucher, GM1 gangliosidosis and STXBP1
- Karyotype in 4 dysmorphic children: 3 Down syndrome (free trisomy, translocation, mosaic); microarray found a 5p/4q change in the fourth

## What it means for doctors

About half the children had a cause that began before birth, and about **1 in 3 had a genetic cause**, most found by exome or genome sequencing. A diagnosis explains the delay, informs recurrence risk for the next pregnancy and, in some conditions, points to specific treatment or monitoring. Birth asphyxia remains an important cause in India, so testing complements, not replaces, the clinical history.

## Testing order (guidelines)

- **No clear cause:** chromosomal microarray + Fragile X first (AAP 2014); yield about 15-20% for microarray.
- **Microarray negative:** exome, ideally trio; ACMG 2021 supports exome or genome as first- or second-tier. Exome yield about 31%, up to 53% with other organ involvement or anomalies (CHOP pathway).
- **Features suggest a specific cause:** karyotype (e.g. Down syndrome); metabolic tests only when clinical clues point to them.

EPISODE 11 · SALES FIELD SHEET

# Delay and intellectual disability: how to approach doctors

| Topic | What to do or say |
|---|---|
| **Doctors to visit** | Paediatricians, paediatric neurologists, developmental paediatricians, neonatologists (NICU follow-up), child psychiatrists (autism); physiotherapy, occupational and speech therapy centres. |
| **Why a cause matters** | Prognosis; recurrence risk for the next child; specific treatment or monitoring in some conditions; fewer repeated tests. |
| **Opening question** | "Doctor, for a child with unexplained delay, what do you send first?" |
| **Facts to quote** | Aurangabad: about 1 in 3 children had a genetic cause. Microarray: 20.9% in 86 children with intellectual disability (Lucknow, Boggula 2015). Clinical exome: 41% in 78 children (Kerala, Valaparambil 2025). |
| **Show** | 30 seconds of Episode 6 (microarray) or Episode 8 (panel or exome); leave the case PDF. |
| **"Too costly"** | Step-wise: microarray first, exome only if needed; singleton first with parental testing later is an option (Episode 3). |
| **"Therapy goes on anyway"** | A cause changes recurrence risk for the family and, in some conditions, treatment or monitoring. |
| **"Parents aren't ready"** | Offer counselling support; testing needs counselling and written consent. |
| **Sample** | EDTA blood for microarray, Fragile X and exome; parents too for trio. Prenatal testing in a later pregnancy: disease only, fetal sex never disclosed (PCPNDT). |
| **Field rules** | Never promise a diagnosis; quote only sourced facts; treatment decisions are the doctor's; confirm price and report time before quoting. |

## References


- Sharma AR et al. Cureus 2023;15(6):e41066. doi.org/10.7759/cureus.41066
- Moeschler JB, Shevell M; AAP Committee on Genetics. Comprehensive evaluation of the child with intellectual disability or global developmental delays. Pediatrics 2014;134(3):e903-e918.
- Manickam K et al. ACMG guideline: exome and genome sequencing for pediatric patients with congenital anomalies or intellectual disability. Genet Med 2021;23:2029-2037.
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- Boggula VR et al. Indian J Med Res 2015;142(6):699-712. Valaparambil KA et al. Indian J Pediatr 2025;92(2):185-190.


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